PROJECT SUMMARY DOCUMENT

Cisco AICTE VIP Program 2026 – AI Track

Field Details :

Project Title NetSage AI – Build an AI Troubleshooting Helper with Human Review

College Name : Lakshmi Narain College of Technology & Science

Technology Track Artificial Intelligence(AI) -

Team Members :

Student Name :	**Enrollment No. :**	**AICTE No. :**	**College :**
Sachin Kumar	**0157AL241156**	STU69d52ee8c76b41775578856	**LNCT&S**
Palak Patil	**0156AL241106**	STU69d28040d25ce1775403072	**LNCT&S**
Pratyush Kothari	**0157AL241123**	STU67936a55152a51737714261	**LNCT&S**
Rohit Barange	**0157AL241145**	STU69d531b3947671775579571	**LNCT&S**

1. Project Title

NetSage AI – Build an AI Troubleshooting Helper with Human Review

NetSage AI is an AI-assisted network troubleshooting, root-cause analysis system aimed at helping network practitioners troubleshoot possible causes of common Cisco-style networking issues. The project combines real-world networking, AI-assisted diagnosis, Python-based technical validation, realistic troubleshooting scenarios, and human review.

The system would take a network troubleshooting scenario, containing symptoms, topology, config, command output, interface status, and connectivity observations, and based on the available information, suggest a possible diagnosis and troubleshooting next steps.

However, the AI diagnosis would always be subjected to further validation, including technical checks via Python-based rules validation and human review, to ensure that the diagnosis is not blindly accepted as correct.

2. Problem Statement

Network troubleshooting poses many challenges, as a single network symptom can have several possible causes. Junior network engineers might know Cisco IOS commands and concepts but struggle to infer multiple symptoms and use the technical evidence at hand to reach a final diagnosis.

For instance, if a PC is unable to reach a server despite receiving an IP address, the cause can range from VLAN, DHCP, DNS, or ACL misconfiguration to NAT, routing, interface, or port issues. A correct diagnosis would require a combination of topology, config, command output, interface status, observations, and an understanding of networking concepts.

NetSage AI aims to provide an AI-assisted approach to network troubleshooting via a structured approach. A set of possible network troubleshooting scenarios would be provided as evidence, and an AI would be used to suggest a possible diagnosis, potentially containing a

- Possible cause,
- OSI layer,
- Supporting technical evidence,
- A command to gather more technical evidence and a recommended troubleshooting next step.

Before final acceptance, such a diagnosis would be subjected to Python-based technical validation and human review.

The project would therefore demonstrate that

AI assistance + technical validation + human review = More reliable network diagnosis

3. Project Overview

NetSage AI is intended to be a comprehensive project centered around AI-assisted network troubleshooting with human review and technical validation. It involves a set of realistic network troubleshooting scenarios with evidence that can be used by an AI to suggest a possible diagnosis, which would then be validated and reviewed by a human.

The project would start with the development of a set of realistic network scenarios containing the relevant technical evidence. Such scenarios would be built using Cisco Packet Tracer, where a network would be configured with specific commands and settings to introduce VLAN misconfigurations, DHCP, DNS, ACL, NAT, routing, connectivity, interface, or port issues.

The AI-assisted diagnosis would then take the relevant technical evidence from such scenarios (topology, config, commands, interfaces, observations) and attempt to provide a possible root-cause diagnosis. The aim would be to help users tie multiple symptoms and pieces of technical evidence to a final root-cause analysis, with a suggested troubleshooting next step.

The suggested diagnosis would then be subjected to Python-based technical validation to ensure that it does not contain any obviously incorrect or unsubstantiated claims. Following that, the diagnosis would go through a human-in-the-loop review to accept, edit, or reject the proposed diagnosis.

The project would involve:

Network scenario / evidence → AI diagnosis → Python validation → Human review → Accepted / Edited / Rejected

as part of the complete workflow.

NetSage AI is intended to be an evidence-based system, where the technical evidence from a network scenario is used to inform the diagnosis and where the diagnosis is always subject to further validation and review.

The major project deliverables would include:

o A structured set of network troubleshooting cases

o Cisco Packet Tracer scenarios and topologies

o AI-assisted diagnosis

o Python rule validation

o Streamlit-based application/dashboard

o Human review

o Testing across different scenarios.

4. Team Contributions and Project Implementation

NetSage AI has been implemented as an end-to-end project that requires collaboration and coordination among all team members. It involved a combination of real-world technical concepts, structured data handling, AI-assisted automation, Python-based rule validation, application development, testing, and human-in-the-loop review, with each team member playing a significant role in the project's completion.

1) Palak Patil – Development Using Python

Palak was responsible for developing the Streamlit application using Python. Using the Streamlit dashboard, the network troubleshooting scenarios, technical evidence, AI diagnosis, and review process can be viewed and tested.

She worked extensively on the code to ensure that the dashboard would be used to demonstrate the various aspects of the project, including the network scenarios, technical evidence, AI diagnosis, Python rule validation, and human review process.

2) Pratyush Kothari – AI Automation Using Python and Grok

Pratyush Kothari implemented the AI automation component using Python and Groq. This component is responsible for taking the structured network scenario and technical evidence and providing an AI-assisted diagnosis.

The AI automation component would take the topology, configuration, troubleshooting commands, interfaces, and observations as input and provide an output containing

o A possible root cause or claim

o Supporting technical evidence

o Applicable OSI layer

o Relevant troubleshooting command

o A suggested troubleshooting next step

o A suggested corrective action, if necessary.

The AI component acts as the advising or recommending entity within the system. Its suggestions would be subjected to both technical validation and human review to ensure accuracy and reliability.

3) Rohit Barange – Packet Tracer Development, Data Handling and Human Review

Rohit Barnage was responsible for developing the Cisco Packet Tracer network scenarios, preparing and organizing the data for the project's scenarios, supporting the testing of the application using real-world network troubleshooting evidence, and participating in the human review process.

He played a critical role in developing each network scenario by designing the topology, configuring the necessary switches, routers, PCs, and other components, and introducing VLAN misconfigurations, DHCP, DNS, ACL, NAT, routing, connectivity, interface, or port issues.

Each scenario contained specific technical evidence that would be used for automation testing, including topology, configuration, commands, interfaces, and observations. Specific areas could also be changed to introduce various network issues and observe their impact on network connectivity and troubleshooting.

In addition, he organized the technical evidence, such as topology, configuration, commands, interfaces, observations, and expected diagnosis, into structured cases. For some of the scenarios, he also participated in the Human-in-the-Loop review process to accept, edit, or reject the proposed diagnosis.

4) Sachin Kumar – Python Rule Checker and Testing:

Sachin Kumar worked on the critical part of the project, the Python rule checker and testing. The rule checker is intended to ensure that any proposed diagnosis from the AI would be subjected to technical scrutiny and be eliminated if it contains unsubstantiated, incorrect, or inconsistent claims.

Sachin participated in testing the application by ensuring that all scenarios had gone through the necessary validation and review steps and that the final diagnosis was always subjected to further scrutiny.

The rule-checking component would highlight many possible diagnosis-related inconsistencies, including

o Diagnosis that does not match the available technical evidence

o Diagnosis that lacks supporting technical evidence

o Diagnosis that contains claims that contradict the available technical evidence

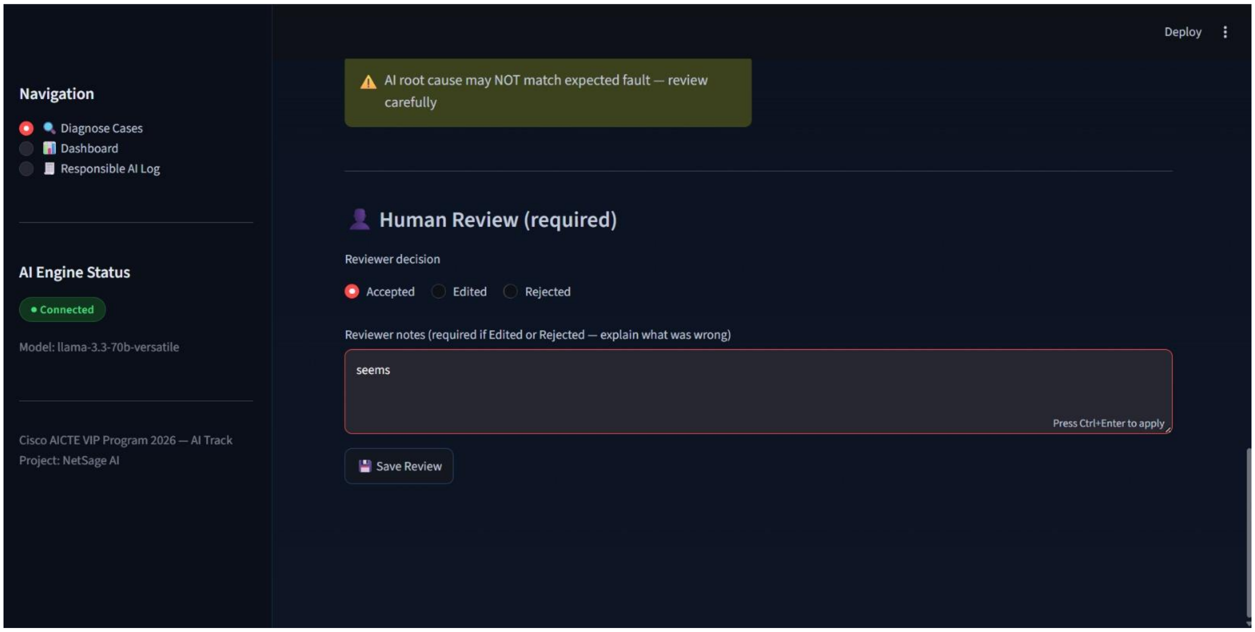
o Unsubstantiated, unsupported, or incorrect diagnosis.

5. Technology Stack Used

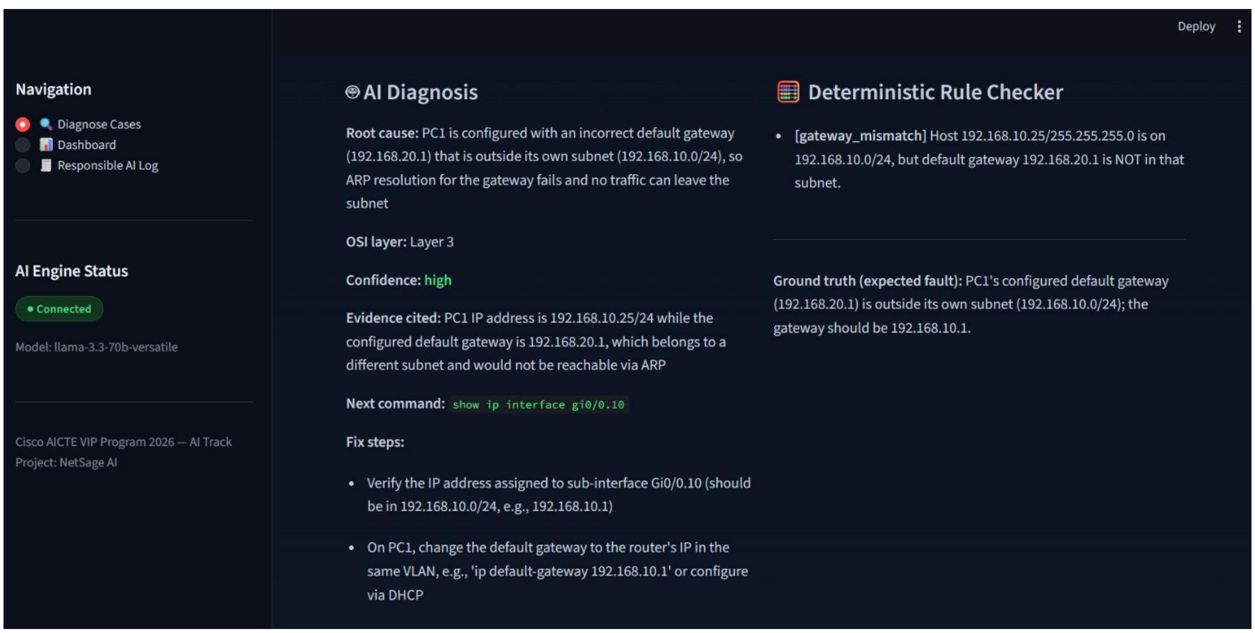
- Cisco Packet Tracer Networking VLAN, DHCP, DNS, ACL, NAT, Routing, Interfaces, Ports
- CSV / JSON Data Structure
- Python AI automation, Rule-checking, Testing, Streamlit App
- Streamlit App Development
- Groq API / LLM AI Automation
- Network Commands Technical Evidence
- Python Rule-Checking Technical Validation
- Human Review Accept / Edit / Reject

6. Some Screenshorts

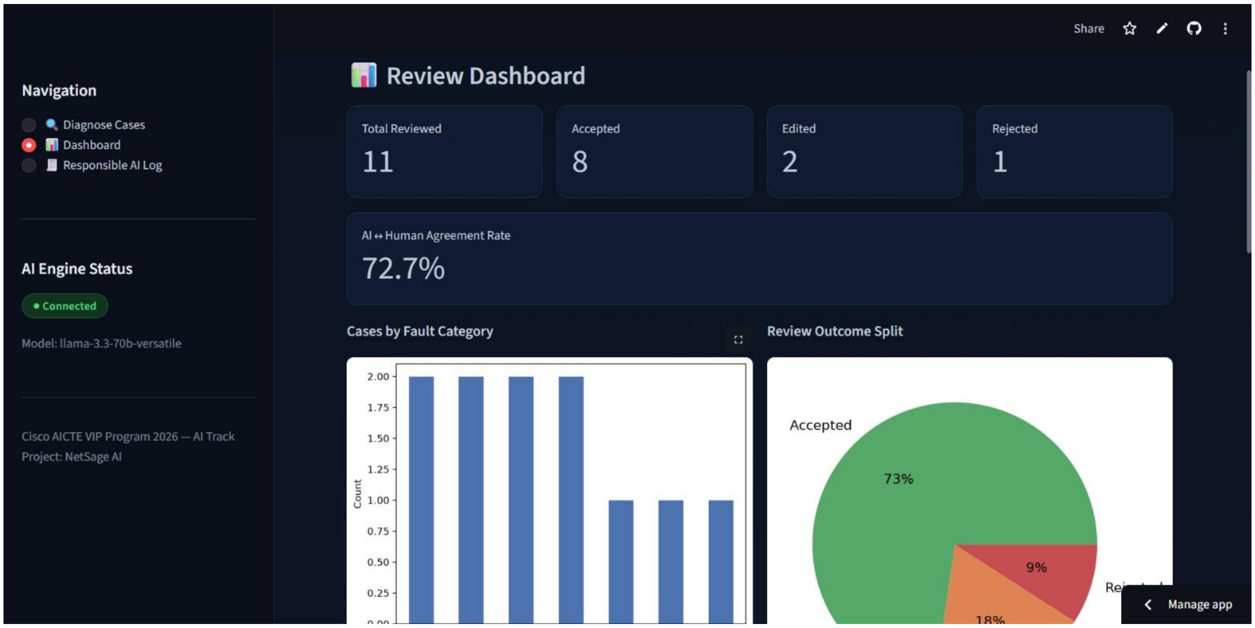
Human review:



Ai Diagnosis:



Result:



7. Responsible AI / Safety Rule Implementation

NetSage AI has been designed to follow Responsible AI guidelines and has implemented several safety rules to prevent misleading, deceptive, or harmful content.

Responsible AI guidelines suggest that an AI model should not be fully relied upon to make decisions or provide responses that have a lasting impact. As such, the AI-generated content should always be reviewed, fact-checked, and validated before being accepted as a reliable source of information. This approach has been implemented in NetSage AI, where an AI diagnosis is subjected to technical validation and human review before being accepted as a reliable suggestion.

The Responsible AI safety rules used in the project include:

AI diagnosis generation → Technical validation → Human review → Accept / Edit / Reject

The technical validation is essential to ensure that the generated diagnosis is not incorrect, unsubstantiated, inconsistent, or otherwise incorrect. Even if the diagnosis has been accepted by a human reviewer, it is always important to remember that an AI model is not fully reliable and should not be used as the sole source of information or decision-making.

By following the Responsible AI guidelines, the NetSage AI system aims to minimize the possibility of providing misleading, deceptive, or harmful information by implementing technical validation and human review for every AI diagnosis.

8. Challenges Faced & How the Team Addressed Them

- Challenge 1: Creating realistic network troubleshooting scenarios Realistic network troubleshooting scenarios are required to implement the project and demonstrate how AI can be used for network troubleshooting. Several scenarios need to be configured using Cisco Packet Tracer to introduce various network connectivity issues. Cisco Packet Tracer was used to design realistic network topologies and scenarios that could be used for troubleshooting and testing.

- Challenge 2: Ensuring that the case and simulation represent the same issue The packet tracer simulation and scenario case need to represent the same underlying issue for testing and validation purposes. The cases needed to be created based on the issues introduced in Packet Tracer to ensure that the same underlying cause is being diagnosed during troubleshooting.

- Challenge 3: Different network issue categories VLAN, DHCP, DNS, ACL, NAT, routing, interfaces, and ports require different levels of troubleshooting. Cases were organized according to the issue category whenever possible and the technical evidence was prepared accordingly.

- Challenge 4: Collecting useful technical evidence A network environment can contain a large body of technical evidence, but not all of it may be relevant or useful for troubleshooting. Cases were created containing the relevant topology, configuration, commands, interfaces, observations, and other technical evidence needed for troubleshooting.

- Challenge 5: Preventing blind acceptance of AI diagnosis An AI diagnosis might appear plausible but contain unsubstantiated, incorrect, or inconsistent claims. The project has implemented Python rule-checking and Human-in-the-Loop review to verify, validate, and fact-check any AI diagnosis before accepting it as reliable information.

- Challenge 6: Testing different levels of complexity Simple network troubleshooting cases might demonstrate the concept, but they fail to represent realistic scenarios and show the full capabilities of the application. The team worked on multiple cases ranging from simple to complex and with various types of issues to highlight the capabilities of the application.

9. What the Team Learned

The team gained significant experience working on the NetSage AI project, including the practical application of networking concepts and troubleshooting, creating realistic scenarios, organizing technical evidence, working with Python code, training, testing, and utilizing an AI model, developing a Streamlit dashboard, implementing technical validation and human review, and learning about Responsible AI rules and guidelines.

Working on the project allowed the team to see the interplay between multiple concepts, including realistic network troubleshooting scenarios, structured data handling, AI automation, Python validation, application development, and human review. The team learned the importance of technical evidence in achieving reliable results when using AI for technical diagnosis. In addition, the team learned that any AI-generated information should always be treated with caution and should undergo technical validation and human review before being accepted as accurate information.

Final Project Summary

NetSage AI – Build an AI Troubleshooting Helper with Human Review is an AI-assisted network trouble-shooting, root-cause analysis project that utilizes real-world network scenarios, structured data handling, AI automation, Python rule-checking, application development, and human review. The project demonstrates how different concepts and components can be used together to achieve better results.

NetSage AI shows how human review, technical validation, and AI assistance can be combined to improve network troubleshooting.

The main focus of the project was on developing realistic network troubleshooting scenarios with technical evidence that can be used to suggest a root-cause diagnosis. Such a diagnosis can then be validated and reviewed by humans to ensure that it is not blindly accepted by users. The main message of the project is that

AI assistance + technical validation + human review = More reliable network diagnosis.